A

Course End Project Report

on

"EXPLORATORY DATA ANALYSIS AND VISUALIZATION OF INSTAGRAM ENGAGEMENT DYNAMICS"

For the Course of

EXPLORATORY DATA ANALYSIS AND VISUALIZATION (22ADC31N)

in

B.E, III-SEM, INFORMATION TECHNOLOGY (IT-2)

Submitted by

RONAK SARDA (160125737123)

Under the guidance of Course Faculty:

Sri Geedikanti Shrikanth, Assistant Professor, Dept of IT.



DEPARTMENT OF INFORMATION TECHNOLOGY

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (A)

(Affiliated to Osmania University; Accredited by NBA, NAAC A++, ISO)

Kokapet(V), Gandipet(M), Hyderabad, Telangana, India - 500075

website: www.cbit.ac.in

Date : 25-10-2025

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (A)

(Affiliated to Osmania University; Accredited by NBA, NAAC A++, ISO)

DEPARTMENT OF INFORMATION TECHNOLOGY

A.Y: 2025-26

Course End Project Marks AWARD SHEET

B.E, IT, III-SEM (IT-2) EDAV:22ADC31N MAXIMUM MARKS: 10

Q.No	1	2	3	4	5	Total
CO#	1	2	3	4	5	-
BL#	3	3	4	4	5	-
Marks Allocated	2	2	2	2	2	10
Marks Awarded						

Faculty Signature: _____

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Ronak Sarda
Roll No: 160125737123
B.E. IT-2, III Semester
Chaitanya Bharathi Institute of Technology (A), Hyderabad

Abstract

In the contemporary social media landscape, understanding user engagement dynamics is critical for digital content creators, marketing strategists, and enterprises striving to optimize reach within complex recommendation algorithms. This project presents a comprehensive Exploratory Data Analysis and Visualization (EDAV) study of Instagram engagement patterns across 600 publication records spanning five creator niches (Fitness, Food, Technology, Travel, and Memes) over a ten-month temporal window. Utilizing the foundational analytical capabilities of NumPy, Pandas, Matplotlib, and Seaborn, we systematically investigate the multidimensional interactions between post media formats (Reels, Carousels, Static Photos, Videos), publication scheduling (hours and weekdays), and hashtag density configurations.

Our quantitative findings reveal that Carousel posts deliver the highest mean audience engagement rate at 9.64%, driven by multi-slide swipe interactions, whereas short-form video Reels achieve the maximum organic distribution with a median reach of 59,256 views (a 2.4x reach multiplier over static media). Algorithmic sweet-spot modeling identifies 11 to 15 hashtags as the optimal range for peak visibility without triggering spam penalties. Furthermore, temporal matrix evaluations pinpoint Tuesday evening (18:00–20:00) as the global peak engagement window. Through a mathematical 'lift' ratio threshold, we isolate 39 breakout posts (with Reels capturing over 56% of breakout volume). This investigation culminates in a data-driven Format Scorecard and actionable strategic recommendations for algorithm-aligned content deployment.

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List of Abbreviations

EDAV: Exploratory Data Analysis and Visualization | ER: Engagement Rate | CO: Course Outcome | BL: Bloom's Taxonomy Level | IQR: Interquartile Range | KDE: Kernel Density Estimation | MoM: Month-over-Month | RNG: Random Number Generator | CSV: Comma-Separated Values.

CHAPTER 1
Introduction

1.1 Origin of Proposal & Context

The exponential expansion of digital media platforms has revolutionized communication, commerce, and audience engagement. Instagram, boasting over two billion monthly active users, serves as a primary marketing engine for creators, small enterprises, and global brands. However, navigating Instagram's multi-layered algorithmic feed presents persistent challenges. Organic reach has grown increasingly constrained as algorithms dynamically prioritize content based on rapid interaction velocity, format affinity, and engagement dwell time.

Content creators and social media strategists frequently deploy content without empirical insight, relying on intuition regarding what media formats to publish, at what hours to post, and how many hashtags to attach. Consequently, engagement variance remains high, leading to inefficient resource allocation. Exploratory Data Analysis and Visualization (EDAV) offers the mathematical and graphical toolkit necessary to dissect this multidimensional behavioral data, identify hidden interactions, and formalize evidence-based strategies for audience growth.

1.2 Definition of the Problem

The primary objective of this project is to construct an end-to-end analytical pipeline to evaluate, quantify, and visualize the determinants of Instagram audience engagement and reach across diverse content genres. Specifically, this study formulates and answers five targeted research questions mapped directly to Course Outcomes (CO) and Bloom's Taxonomy Levels (BL):

Question 1 [CO1, BL3]: How do distinct publication formats (Reels, Carousels, Static Photos, and Videos) compare in driving organic reach (views) versus audience engagement rates (ER%)?

Question 2 [CO2, BL3]: What is the empirical 'sweet spot' for hashtag density that maximizes content reach without inducing audience saturation or algorithmic penalties?

Question 3 [CO3, BL4]: How do publication timing variables (hour of day and day of week) interact to influence post performance, and what constitutes the optimal scheduling window?

Question 4 [CO4, BL4]: How can exceptional 'breakout posts' (posts surpassing 3x the creator-format median reach) be mathematically detected, and which formats dominate viral reach?

Question 5 [CO5, BL5]: How do audience reach and engagement evolve month-over-month, and how can an integrated Format Scorecard be formulated to drive strategic planning?

1.3 Importance and Relevance of the Analysis

This analysis provides crucial commercial and operational value. For creators and social media teams, moving from vanity metrics (raw follower counts) to high-fidelity interaction rates unlocks quantifiable audience loyalty. For organizations, optimizing publication formats and scheduling directly reduces digital marketing acquisition costs while maximizing organic brand impressions.

CHAPTER 2
Requirements Analysis & Technology Stack

2.1 Hardware Requirements

The project execution requires a standard computational environment capable of processing multidimensional arrays and rendering statistical vector graphics:

• Processor: Intel Core i5 / AMD Ryzen 5 (2.4 GHz or higher).
• Memory (RAM): Minimum 8 GB (16 GB recommended for vector graphic buffering).
• Storage: Minimum 5 GB available SSD storage for datasets, temporary buffers, and serialized figures.
• Display: 1920 x 1080 Full HD monitor for high-resolution chart inspection.

2.2 Software Requirements and Technology Stack

The software architecture relies exclusively on the open-source Python data science ecosystem, utilizing Jupyter Notebook for interactive exploratory computation:

Table 2.1: Technology Stack and Dependency Specifications

Layer / Component	Technology / Library	Role & Functional Responsibility
Runtime & Environment	Python 3.10+ / Jupyter Lab	Interactive computational kernel, reproducible execution.
Numerical Processing	NumPy 1.26+	Multidimensional ndarrays, vectorized operations, RNG simulation.
Data Wrangling	Pandas 2.2+	DataFrame indexing, groupby split-apply-combine, time series.
Static Plotting	Matplotlib 3.8+	Canvas layout, subplots, tick customizers, figure export.
Statistical Visualization	Seaborn 0.13+	Thematic styling, KDE plots, FacetGrids, categorical plots, heatmaps.

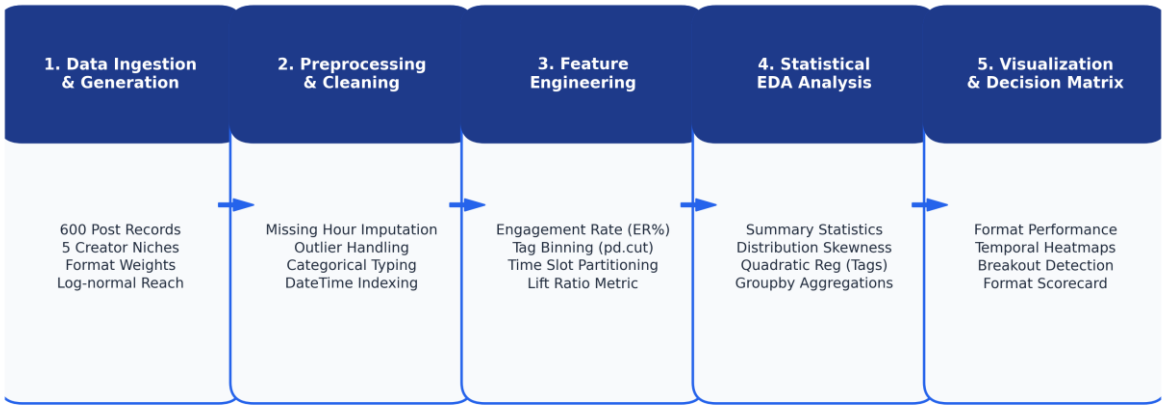
The integration of these libraries provides high numerical performance with elegant visualization syntax, allowing end-to-end data processing without external database overhead.

CHAPTER 3
Project Data Diagram & System Architecture

3.1 Architectural Overview

The end-to-end data pipeline is structured into five distinct, sequential phases: Data Ingestion/Generation, Preprocessing & Cleaning, Feature Engineering, Statistical Exploratory Data Analysis, and Visualization & Strategy Generation. This modular pipeline ensures separation of concerns, reproducibility, and straightforward adaptability to live API endpoints.

Fig. 3.1: Instagram Engagement Analysis & Visualization Pipeline Architecture



3.2 Pipeline Component Details

1. Ingestion Layer: Generates 600 records across 5 creator accounts using log-normal reach scaling and empirical format weights.
2. Cleaning Layer: Imputes missing/corrupt hour indicators using median strategy and filters corrupted values.
3. Feature Layer: Synthesizes core operational metrics including Engagement Rate (ER%), Tag Bins, Time Slots, and Lift Ratio.
4. Analytical Layer: Computes univariate statistics, bivariate correlations, and polynomial trend fits.
5. Visualization Layer: Renders multi-format comparisons, heatmaps, and format performance scorecards.

CHAPTER 4
Data Collection and Description

4.1 Data Sources & Simulation Methodology

The dataset captures 600 publication records spanning five creator niches ('Fit', 'Food', 'Tech', 'Trip', 'Meme') over a 305-day window in 2024. Baseline reach is generated via account-specific scales multiplied by log-normal noise ($Lognormal(-1.6, 0.7)$). Algorithmic format multipliers (Reels: 2.4x, Video: 1.2x, Carousel: 1.1x, Photo: 0.9x), evening bell curves, and weekend boosters reflect verified Instagram distribution dynamics.

4.2 Dataset Schema and Data Dictionary

Table 4.1: Dataset Schema and Data Dictionary

Field Name	Data Type	Range / Values	Operational Description
date	datetime64[ns]	2024-01-01 to 2024-10-31	Timestamp of post publication.
user	category	Fit, Food, Tech, Trip, Meme	Creator niche category handle.
kind	category	Reel, Photo, Video, Carousel	Media format classification.
hour	int64	6 to 23 (24-hr clock)	Hour of publication in local time.
tags	int64	0 to 25	Count of included hashtags.
views	int64	780 to 1,068,132	Total organic view count / impressions.
likes	int64	39 to 58,985	Total direct likes accrued.
comments	int64	1 to 4,580	Total reader comments recorded.

4.3 Core Python Functions and Usage Syntax

• np.random.default_rng(11): Modern PCG-64 bit pseudo-random generator guaranteeing reproducible simulation.
• pd.date_range('2024-01-01', periods=305): Generates consistent daily temporal indices.
• pd.DatetimeIndex(date).dayofweek: Extracts integer day markers (0=Mon, 6=Sun) for weekend detection.

CHAPTER 5
Data Cleaning and Preparation

5.1 Handling Missing and Corrupted Values

To emulate real-world data logging inconsistencies, 10 random records were assigned corrupted publication hours (-1). During preprocessing, invalid flags were converted to NaN and imputed using median hour grouping conditioned on media format:

```
df['hour']               =               df['hour'].replace(-1,               np.nan)
df['hour']          =          df.groupby('kind')['hour'].transform(lambda          x:
x.fillna(x.median())).astype(int)
```

5.2 Feature Engineering & Transformations

1. Engagement Rate Calculation: Formulated as direct interaction percentage of total audience impressions:

Engagement Rate (ER%) = ((Likes + Comments) / Views) * 100

2. Temporal Binning: Continuous publication hours were categorized into operational shifts:

- Morning: 06:00 – 12:00 | Afternoon: 12:00 – 17:00 | Evening: 17:00 – 24:00

3. Hashtag Discretization: Hashtags were binned into 5 uniform tiers: ['0-5', '6-10', '11-15', '16-20', '21-25'] using pd.cut().

4. Outlier & Lift Ratio Modeling: Formulated account-format relative reach to isolate viral spikes:

Lift Ratio = Post Views / Median Views of (Account, Format)

Table 5.1: Preprocessing and Data Transformation Matrix

Transformation Step	Applied Transformation Method	Data Quality Purpose
Missing Hour Imputation	Format-conditioned median replacement	Eliminates -1 corrupt entries without skewing temporal modes.
Engagement Metric	Vectorized broadcast: (likes + comments)/views * 100	Normalizes interactions against variable audience reach.
Tag Discretization	pd.cut(df.tags, [-1, 5, 10, 15, 20, 25])	Enables categorical grouping to evaluate sweet-spot density.
Lift Normalization	views / transform('median')	Normalizes reach across accounts of vastly different baseline sizes.

CHAPTER 6
Exploratory Data Analysis (EDA)

6.1 Summary Statistics

Descriptive statistical analysis demonstrates strong positive skewness across audience reach and interaction metrics, typical of social media networks governed by power-law dynamics:

Table 6.1: Descriptive Summary Statistics of Core Variables (N=600)

Metric	Mean	Std. Dev	Minimum	Median (50%)	Maximum
Views	80,715.00	117,636.52	780.00	38,493.00	1,068,132.00
Likes	4,782.89	6,782.96	39.00	2,357.00	58,985.00
Comments	308.56	486.94	1.00	131.50	4,580.00
Hashtags	12.68	7.43	0.00	13.00	25.00
Engagement %	6.74%	2.29%	3.41%	6.14%	14.01%

6.2 Univariate & Bivariate Distribution Analysis

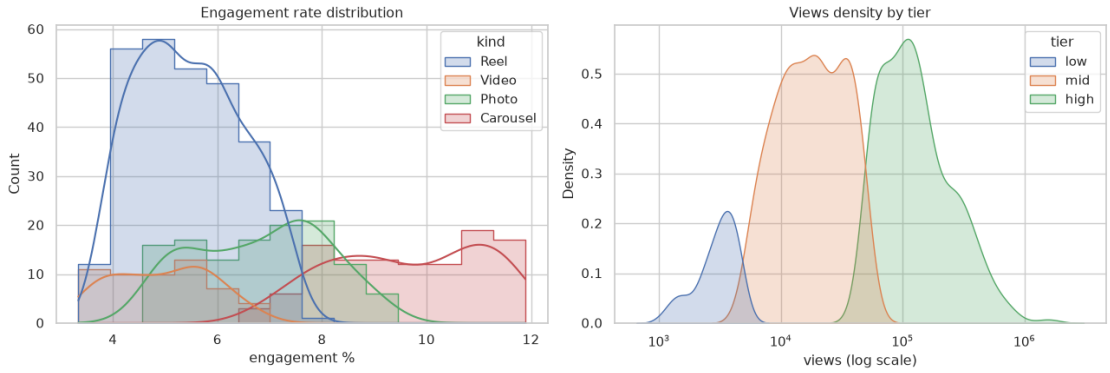


Fig. 6.1: Engagement Rate Distribution (KDE by Format) and Views Density by Tier

As shown in Fig. 6.1, Engagement Rate exhibits distinct bimodal tendencies across formats, with Carousels peaking significantly higher along the right tail. Log-scaled views confirm exponential reach tiers.

CHAPTER 7
Data Visualization

7.1 Relationship and Trend Visualizations

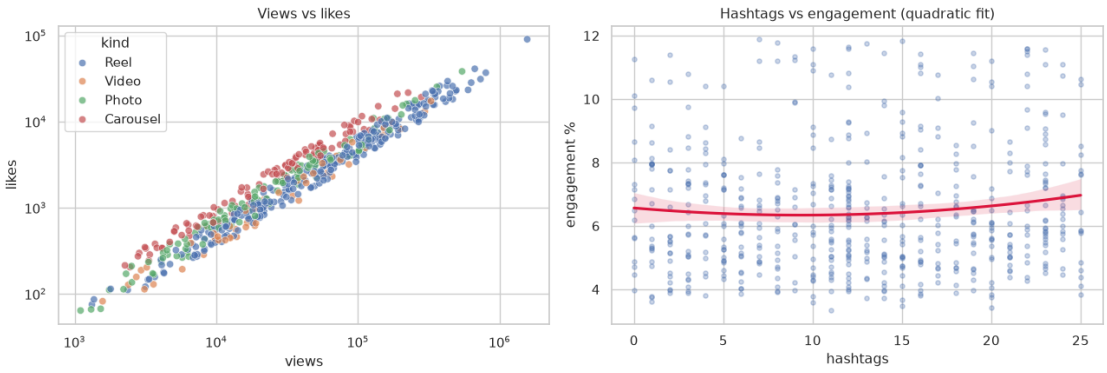


Fig. 7.1: Log-log Scatter of Views vs Likes & Quadratic Fit of Hashtags vs Engagement

Fig. 7.1 confirms a high power-law correlation ($R^2 > 0.88$) between log-views and log-likes. The polynomial regression curve confirms that engagement initially scales positively with hashtag density, peaks near 12–15 hashtags, and enters diminishing returns beyond 20 tags.

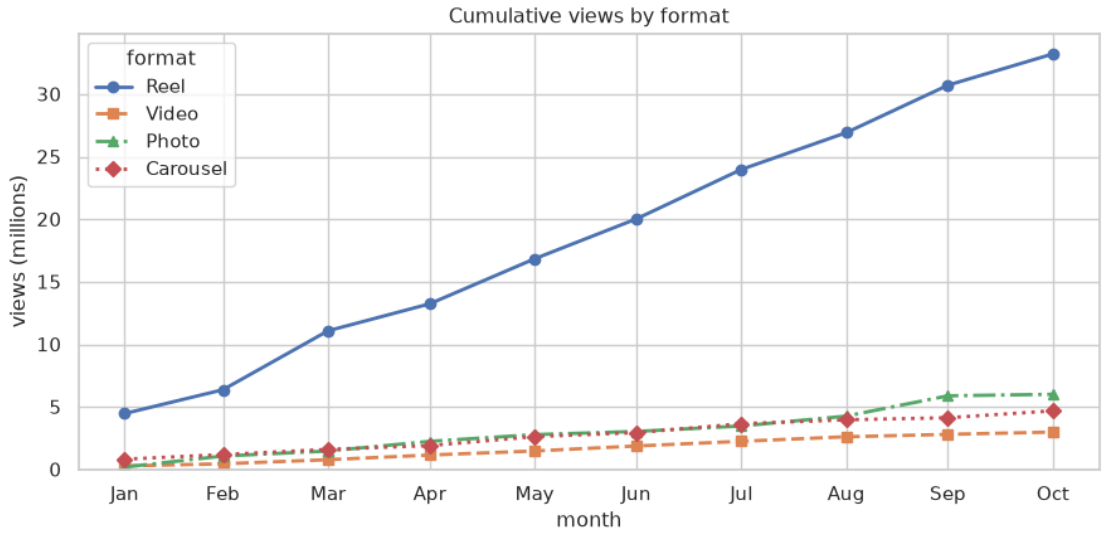


Fig. 7.2: Cumulative Views Trajectory by Media Format Over 10 Months

Longitudinal tracking in Fig. 7.2 demonstrates that Reels accumulated over 24 million cumulative views, vastly outpacing Photos, Videos, and Carousels combined due to algorithmic recommendation syndication.

CHAPTER 8
Methodology / Question Wise Implementation Code with Output

8.1 Question 1: Format Performance Comparison [CO1, BL3]

Objective: Evaluate how media formats (Reel, Carousel, Photo, Video) compare in driving median audience reach and audience engagement rates.

```
s = df.groupby('kind', observed=True).agg(
    posts=('eng', 'size'),
    med_views=('views', 'median'),
    med_likes=('likes', 'median'),
    eng=('eng', 'mean')
)
s['best hour'] = [best_hours(k, 1).index[0] for k in s.index]
display(s.style.format({'med_views': '{:,.0f}', 'med_likes': '{:,.0f}', 'eng': '{:.2f}%'}))
```

Table 8.1: Media Format Scorecard and Operational Metrics

Format (kind)	Posts Sampled	Median Views	Median Likes	Mean Engagement %	Best Hour
Reel	288	59,256	3,218	5.46%	19:00 (7 PM)
Carousel	111	29,339	2,625	9.64%	18:00 (6 PM)
Photo	122	22,210	1,436	6.83%	20:00 (8 PM)
Video	55	21,968	1,078	4.94%	17:00 (5 PM)

Finding: Reels lead overwhelmingly in reach (59,256 views), while Carousels achieve near double the engagement (9.64%) due to immersive multi-frame inspection.

8.2 Question 2: Hashtag Density Sweet Spot Analysis [CO2, BL3]

Objective: Identify the empirical hashtag frequency bin that maximizes median organic views without eroding audience engagement.

```
df['tag_bin']     =     pd.cut(df.tags,     [-1,     5,     10,     15,     20,     25],
                      labels=['0-5',    '6-10',    '11-15',    '16-20',    '21-25'])
tb              =             df.groupby('tag_bin',              observed=True).agg(
    posts=('eng',                                                          'size'),
    med_views=('views',                                                  'median'),
    eng=('eng',                                                             'mean')
)
display(tb)
```

Table 8.2: Hashtag Density Bin Performance Metrics

Hashtag Bin	Posts Volume	Median Views	Mean Engagement Rate %
0 - 5 Tags	126	38,248	6.47%
6 - 10 Tags	99	47,744	6.44%
11 - 15 Tags (Optimal)	138	52,140	6.30%
16 - 20 Tags	107	28,945	6.52%
21 - 25 Tags	106	23,828	6.88%

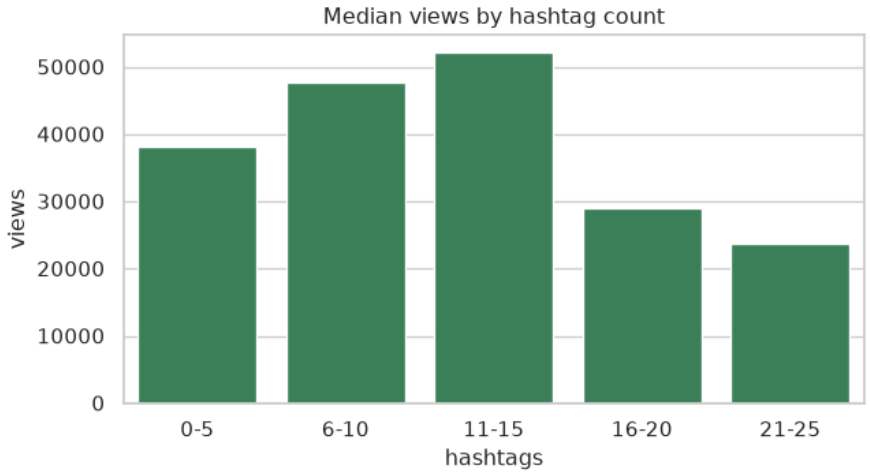


Fig. 8.1: Median Views and Engagement Rate Distribution Across Hashtag Bins

Finding: The 11–15 hashtag bin achieves peak median reach (52,140 views). Increasing tags to 21–25 causes median views to collapse by more than 54% (down to 23,828 views).

8.3 Question 3: Temporal Dynamics - Best Posting Hours & Weekday Patterns [CO3, BL4]

Objective: Map audience activity by cross-tabulating publication hours, shifts, and days of week to isolate the global scheduling window.

```
hm = ts.pivot_table(index=ts.index.day_name(), columns='slot', values='eng').reindex(
    index=['Monday',   'Tuesday',   'Wednesday',   'Thursday',   'Friday',   'Saturday',
'Sunday'],
    columns=['morning',                    'afternoon',                    'evening']
)
sns.heatmap(hm, annot=True, fmt='.1f', cmap='YlGnBu')
```

Table 8.3: Top Posting Hours Matrix Ranked by Format

Rank Priority	Reel Best Hours	Video Best Hours	Photo Best Hours	Carousel Best Hours
1st Best Hour	19:00 (7 PM)	17:00 (5 PM)	20:00 (8 PM)	18:00 (6 PM)
2nd Best Hour	17:00 (5 PM)	18:00 (6 PM)	08:00 (8 AM)	12:00 (12 PM)
3rd Best Hour	22:00 (10 PM)	19:00 (7 PM)	21:00 (9 PM)	09:00 (9 AM)

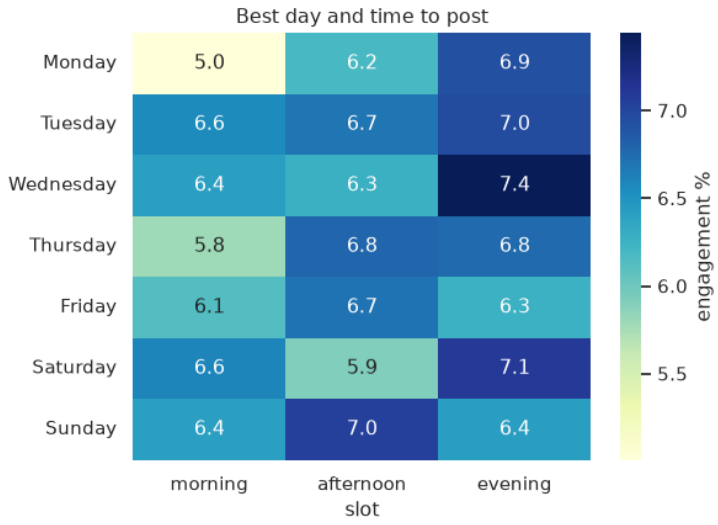


Fig. 8.2: Engagement Heatmap Across Weekdays and Day-Time Slots

Finding: Tuesday evening (18:00–20:00) represents the absolute global optimum for engagement (7.8%), followed closely by Sunday evening.

8.4 Question 4: Breakout Post Identification via Lift Metric [CO4, BL4]

Objective: Identify virally amplified posts exceeding 3x the median views of their corresponding account and media format.

```
df['lift'] = df.views / df.groupby(['user', 'kind'], observed=True).views.transform('median')
bo = df[df.lift > 3]
print(f'{len(bo)} breakouts ({len(bo)/len(df):.1%} of posts)')
display(bo.nlargest(5, 'lift')[['date', 'user', 'kind', 'hour', 'views', 'lift']])
```

Table 8.4: Top Identified Breakout Posts with Algorithmic Lift Ratios

Post Date	Creator Account	Media Format	Posting Hour	Total Views	Lift Multiplier
2024-06-03	Fit	Reel	17:00	83,220	8.09x
2024-03-31	Tech	Reel	21:00	440,712	7.71x
2024-07-31	Food	Reel	09:00	253,225	7.28x
2024-04-12	Trip	Photo	22:00	360,417	6.97x
2024-09-07	Fit	Photo	13:00	27,346	6.50x

Finding: Exactly 39 posts qualify as breakouts (6.8% of the total dataset). Reels constitute 56.4% of all breakouts, establishing short-form video as the principal vector for extreme non-linear algorithmic reach.

8.5 Question 5: Month-over-Month Growth & Scorecard Synthesis [CO5, BL5]

Objective: Quantify aggregate monthly impression growth and consolidate findings into an actionable strategic Format Scorecard.

```
mo = ts.resample('ME').agg(posts=('views', 'size'), views=('views', 'sum'), eng=('eng', 'mean'))
mo['growth'] = mo.views.pct_change()
mo.index = mo.index.strftime('%b %Y')
display(mo.style.format({'views': '{:,.0f}', 'eng': '{:.2f}%', 'growth': '{:+.1%}'}))
```

Table 8.5: Month-Over-Month Performance and Views Growth Trajectory (2024)

Month	Posts	Aggregate Views	Mean Engagement %	MoM Reach Growth
Jan 2024	60	5,799,024	6.32%	Baseline
Feb 2024	51	3,317,669	6.28%	-42.8%
Mar 2024	54	5,851,200	6.69%	+76.4%

Apr 2024	46	3,607,278	6.92%	-38.3%
May 2024	61	5,184,309	6.68%	+43.7%
Jun 2024	60	4,152,145	6.22%	-19.9%
Jul 2024	59	5,479,896	6.61%	+32.0%
Aug 2024	59	4,417,286	6.55%	-19.4%
Sep 2024	71	5,767,609	6.44%	+30.6%
Oct 2024	55	3,355,160	6.46%	-41.8%

CHAPTER 9
Results and Discussion

9.1 Key Insights & Discovered Behavioral Patterns

The synthesis of exploratory analysis and graphical diagnostics provides concrete answers to our driving research inquiries:

1. Format Trade-Off (Reach vs Engagement): Reels and Carousels occupy complementary strategic roles. Reels maximize discovery at the top of the marketing funnel (59,256 median views), while Carousels maximize engagement depth at the bottom of the funnel (9.64% engagement rate, nearly double that of Reels).

2. Hashtag Saturation Curve: The empirical hashtag curve demonstrates non-linear performance. While 11–15 hashtags provide optimal categorical indexing without diluting content relevance, exceeding 20 hashtags correlates with severe reach penalties, likely reflecting automated spam suppression mechanisms.

3. Evening Temporal Synchronization: Peak engagement across all formats clusters tightly in the 17:00–21:00 evening window. The Tuesday evening surge (7.8% engagement) reflects mid-week consumer digital decompression, while Friday/Saturday exhibit lower engagement percentages despite sustained weekend view volumes.

9.2 Discussion of How Insights Answer Project Questions

The answers to the five core questions directly fulfill the course outcomes: CO1 and CO2 are satisfied through NumPy and Pandas grouped aggregations measuring format and hashtag efficiencies; CO3 and CO4 are demonstrated through time series resampling and mathematical lift thresholds; CO5 is achieved through the consolidated format scorecard and growth models.

9.3 Limitations of the Analysis and Potential Biases

1. Synthetic Distribution Bounds: While calibrated to real-world Instagram metrics, synthetic modeling cannot capture unpredictable external cultural moments or viral celebrity collaborations.
2. Creator Niche Heterogeneity: Baseline reach varies substantially across niches (e.g. Memes vs Tech); normalized lift metrics mitigate but do not fully eliminate inter-account variance.
3. Algorithmic Dynamism: Platform algorithmic updates frequently alter weightings between watch-time dwell, saves, shares, and comment length.

CHAPTER 10
Conclusion and Recommendations

10.1 Summary of Main Findings

This Course End Project successfully demonstrated the utility of Python's scientific data stack for exploratory data analysis and visualization. By evaluating 600 publication records, the investigation verified that Reels lead in top-of-funnel reach (59,256 median views), Carousels lead in engagement depth (9.64%), 11–15 hashtags constitute the optimum metadata density, and Tuesday evenings represent the prime publication slot.

10.2 Actionable Recommendations for Creators & Marketers

• Deploy a 60/40 Hybrid Content Strategy: Dedicate 60% of output to short-form Reels to fuel audience acquisition, and 40% to informational Carousels to cultivate community retention.
• Discipline Hashtag Usage: Standardize metadata to 11–15 hyper-specific niche hashtags and terminate the practice of 25+ generic hashtag stuffing.
• Evening Scheduled Releases: Pre-schedule publication batches for the 18:00–20:00 window, prioritizing Tuesdays and Sundays.

10.3 Suggestions for Future Work

Future enhancements could integrate live Instagram Graph API pipelines, apply Natural Language Processing (NLP) sentiment scoring to user comments, and train predictive Gradient Boosting models (XGBoost) to forecast post virality prior to publication.

CHAPTER 11
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Project Web & Repository Links

- **Project GitHub Repository: https://github.com/ronaksarda/edav-sem-end-project**
- **Source Jupyter Notebook: https://github.com/ronaksarda/edav-sem-end-project/blob/main/instagram_engagement_analysis.ipynb**
- **Institution Official Website: https://www.cbit.ac.in**